

CONTROLLED UNCLASSIFIED INFORMATION (CUI)

Secure File Encryption for Email Transmission

Before You Begin: Choose the Right Method

PREFERRED: S/MIME Encrypted Email

If you have your recipient's S/MIME certificate (PIV, CAC, or other), use S/MIME:

1. Compose your email in Outlook with CUI files attached
2. Click **Sign** (proves it's from you) and **Encrypt** (only they can read it)
3. The email is automatically secured end-to-end – no password needed

Use SendCUIEmail when:

- You don't have the recipient's S/MIME certificate
- Your recipient doesn't have PKI capability (no PIV/CAC/S/MIME)
- You need to send CUI to external partners or contractors

TIP: Even when using SendCUIEmail, **sign your email** if you have a PIV/CAC. This proves the email came from you (authenticity) even though the recipient can't receive S/MIME encrypted mail.

Part A: Encrypting Files (Sender)

Step 1: Launch the Tool

1. Navigate to the SendCUIEmail folder
2. Double-click `Encrypt.bat`
3. A blue PowerShell window will open

Step 2: Select Files

1. When prompted, drag files into the window OR type the file path
2. Press Enter when done adding files

Step 3: Choose CUI Category

1. Select the appropriate CUI marking (0-4)
2. Press Enter for basic CUI if unsure

Step 4: Create Password

1. Type a password (minimum 8 characters)
2. **IMPORTANT:** Characters will NOT appear as you type – this is normal!
3. Type the password again to confirm

4. Remember this password – you'll share it separately

Step 5: Send the Files

1. Find the output folder (path shown in yellow)
2. Attach the `.Locked` files and `README.md` to your email
3. If you have a PIV/CAC, click **Sign** before sending (proves it's from you)
4. Send the email to your recipient

WARNING: Password Delivery

COMPLIANT: Phone call, SMS text, or in-person

NOT COMPLIANT: Email (even if separate from files)

Part B: Decrypting Files (Recipient)

Step 1: Save the Files

1. Download all .Locked files from the email
2. Download the README.md file
3. Save them all to the same folder (e.g., Desktop)

Step 2: Open PowerShell

1. Press **Windows Key + X**
2. Click “Windows PowerShell” or “Terminal”
3. A blue command window will open

Step 3: Run the Decryption Command

1. Open the README.md file (double-click or use Notepad)
2. Find the “PowerShell One-Liner” section
3. Copy the entire command (starts with `$f=Read-Host`)
4. Paste it into PowerShell and press Enter

Step 4: Enter File Path and Password

1. When prompted for “File”, type the full path to a .Locked file
Example: C:\Users\YourName\Desktop\Document.pdf.Locked
2. When prompted for “Password”, type the password you received
3. **IMPORTANT:** Characters will NOT appear – this is normal!

Step 5: Verify Decryption

1. The decrypted file appears in the same folder
2. *Example: Document.pdf.Locked becomes Document.pdf*
3. Repeat steps 3-4 for each .Locked file

TIP: Use the same password for all files in a batch.
If decryption fails, double-check the password and file path.